École CentraleSupélec

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PIMC simulation of a proton in a one-dimensional double-well potential

Equilibrium distributions, effective barriers, and deterministic numerical checks

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Chapter 1

Introduction

Quantum tunneling and spatial delocalization influence the equilibrium behavior of light particles such as protons. In this project, a proton in a symmetric double-well potential is represented by a classical ring polymer and sampled with Path Integral Monte Carlo (PIMC). This equilibrium mapping incorporates quantum statistical effects through the imaginary-time path integral; it does not simulate a real-time tunneling trajectory [1, 2].

We study two values of the dimensionless quantum parameter, $C = 0.3$ and 0.9, and two reduced temperatures, $\theta = k_B T/V_0 = 0.4$ and 0.6. The measured observables are the mean physical potential energy, the bead position density, and the centroid density. The centroid density defines an effective free-energy profile whose central barrier is compared with the bare barrier V_0.

The Monte Carlo results in this report are the existing saved experiments in `main.ipynb`. A final notebook section now supplies separate deterministic calculations: classical quadrature, a finite-difference solution of the Schrödinger equation, and a transfer-matrix evaluation of the finite-P path integral. These calculations quantify discretization effects and provide an independent check of MC sampling. The evidence supports the physical trends, while finite-P bias and incomplete statistical convergence limit the precision of the reported barriers.

Chapter 2

Theoretical model and numerical method

2.1 Potential and physical scales

The proton, of mass $m = m_p$, experiences

$$V(x) = V_0 \left[\left(\frac{x}{a} \right)^2 - 1 \right]^2, \qquad a = 0.04\,\text{nm}. \tag{2.1}$$

Its minima are at $x = \pm a$, and $V(0) = V_0$. Introduce $y = x/a$, $v(y) = V(x)/V_0 = (y^2 - 1)^2$, and

$$C = \frac{\hbar^2}{2ma^2V_0}, \qquad \theta = \frac{k_B T}{V_0}, \qquad V_0 = \frac{\hbar^2}{2ma^2 C}. \tag{2.2}$$

The code evaluates energies and positions in SI units. Reduced units are used here for interpretation, plotting, and deterministic references.

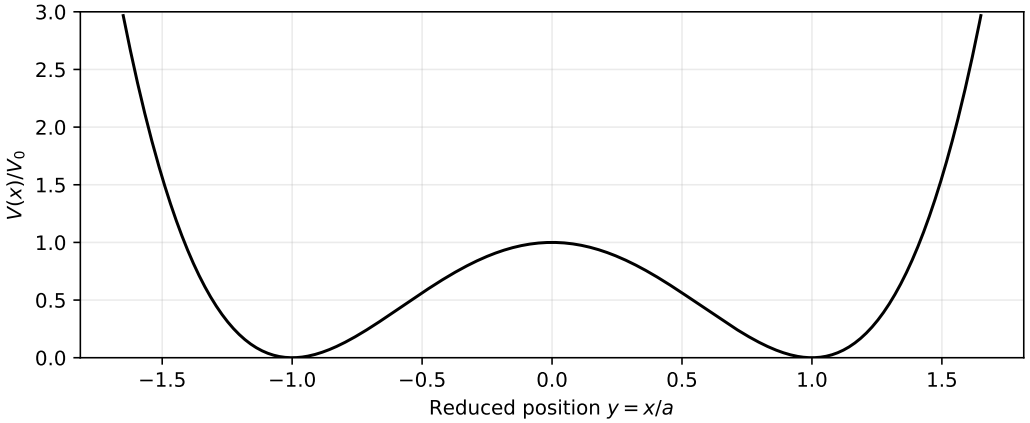


Figure 2.1: The reduced potential used in both numerical methods.

Changing C at fixed m and a changes V_0 and therefore the physical temperature corresponding to a fixed θ:

C	V_0 (eV)	T at $\theta = 0.4$ (K)	T at $\theta = 0.6$ (K)
0.3	0.043229	200.66	300.99
0.9	0.014410	66.89	100.33

The two values of C lie on opposite sides of the critical value $C_0 \simeq 0.732$ given in the project brief, where the ground-state total energy equals V_0 [1]. The deterministic spectrum below confirms this distinction.

2.2 Ring-polymer action and Metropolis moves

For P beads, the effective energy sampled at inverse temperature $\beta = (k_B T)^{-1}$ is

$$U_{\text{eff}} = \sum_{s=1}^{P} \left[\frac{1}{2} m\omega_P^2 (x_{s+1} - x_s)^2 + \frac{V(x_s)}{P} \right], \quad \omega_P = \frac{\sqrt{P} k_B T}{\hbar}, \quad x_{P+1} = x_1. \tag{2.3}$$

Equivalently,

$$\beta U_{\text{eff}} = \sum_{s=1}^{P} \left[\frac{P\theta}{4C} (y_{s+1} - y_s)^2 + \frac{v(y_s)}{P\theta} \right]. \tag{2.4}$$

A bead is selected uniformly at random and displaced by a uniform increment in $[-dx, dx]$. The move is accepted with probability $\min(1, e^{-\beta \Delta U_{\text{eff}}})$. Only the selected bead's potential and adjacent springs enter the energy difference. For $P = 1$ the spring term vanishes. Initially all beads are at $x_s = a$.

One sweep comprises P attempted moves with bead selection with replacement. Rejected moves remain part of the sampled trajectory. Histograms are recorded once per completed production sweep. Energy and centroid arrays are stored after every attempted move; an energy average after B discarded sweeps therefore starts at array index BP. At $P = 15$, 500,000 sweeps correspond to 7.5 million attempted moves.

2.3 Observables and density convention

The physical potential-energy estimator and centroid are

$$\langle V \rangle = \left\langle \frac{1}{P} \sum_s V(x_s) \right\rangle, \qquad y_c = \frac{1}{P} \sum_s y_s. \tag{2.5}$$

All beads contribute to the bead histogram; each configuration contributes one centroid. Both histograms use 100 bins over $y \in [-2.5, 2.5]$ with width $\Delta y = 0.05$. Their densities satisfy $\sum_i \rho_i \Delta y = 1$ over this retained interval. The physical density is $\rho_x(x) = \rho_y(x/a)/a$. This common scale cancels in density ratios used for free energies.

The bead density approximates the quantum position density as P increases. The centroid density describes the average position of an imaginary-time path, a distinct observable. Neither the P beads within a configuration nor successive MC configurations should be assumed statistically independent.

Chapter 3

Classical validation and sampling protocol

3.1 Classical reference and displacement calibration

At $P = 1$, the target density and potential energy are

$$\rho_{\mathrm{cl}}(y) = \frac{e^{-v(y)/\theta}}{\int e^{-v(y)/\theta}\, dy}, \qquad \langle v \rangle_{\mathrm{cl}} = \int v(y)\rho_{\mathrm{cl}}(y)\, dy. \tag{3.1}$$

Question 3 evaluates these integrals with the trapezoidal rule on 2,001 points in $[-2.5, 2.5]$. At $\theta = 0.4$ this gives $\langle V \rangle_{\mathrm{cl}}/V_0 = 0.228637092$. Refinement to 4,001 points on $[-3, 3]$ leaves the displayed value unchanged.

The tested maximum displacements are $dx/a = 0.2, 0.4, 0.8, 1.0, 1.2, 1.6, 2.0, 2.4, 3.0$. Each pilot uses 20,000 attempted moves, of which the first 5,000 are excluded from the acceptance and crossing diagnostics. The value closest to 45% acceptance is $dx = a$, with measured acceptance 44.7%. This is a practical calibration, rather than a proof of optimal sampling efficiency: larger proposed moves produce more well crossings in the pilot.

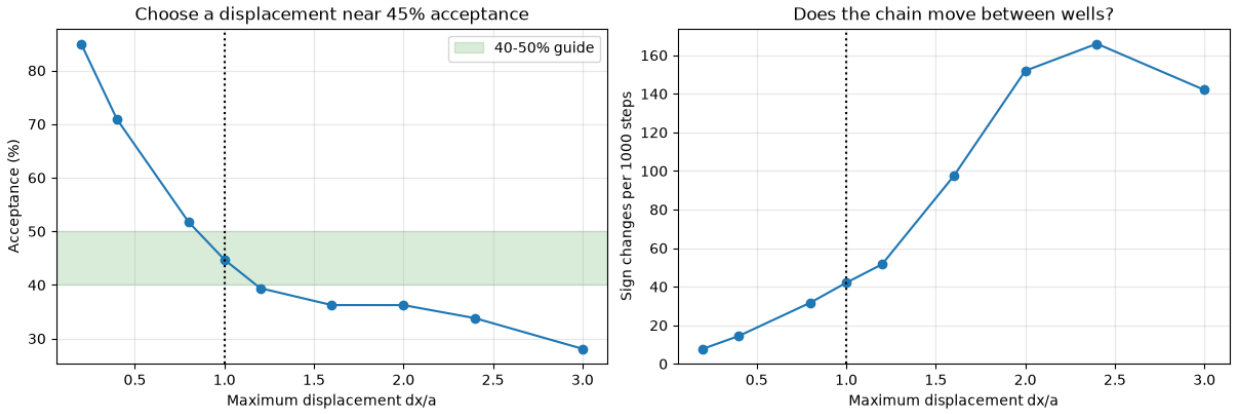


Figure 3.1: Saved classical displacement calibration. The sign-change count is a rough mixing diagnostic and includes rapid recrossings.

Three subsequent classical runs use 200,000 attempted moves each and discard 5,000. Their saved summaries are:

Seed	$\langle x\rangle/a$	$\langle V\rangle/V_0$	Fraction in right well
10	-0.0110	0.2292	0.494
20	-0.0004	0.2268	0.500
30	-0.0256	0.2276	0.486

The means and histogram support agreement with the classical reference. The green bands in Figure 3.2 represent chosen accuracy goals ($|\langle x\rangle| < 0.05a$ and energy within 5% of the reference), not confidence intervals. No block-based standard error is computed.

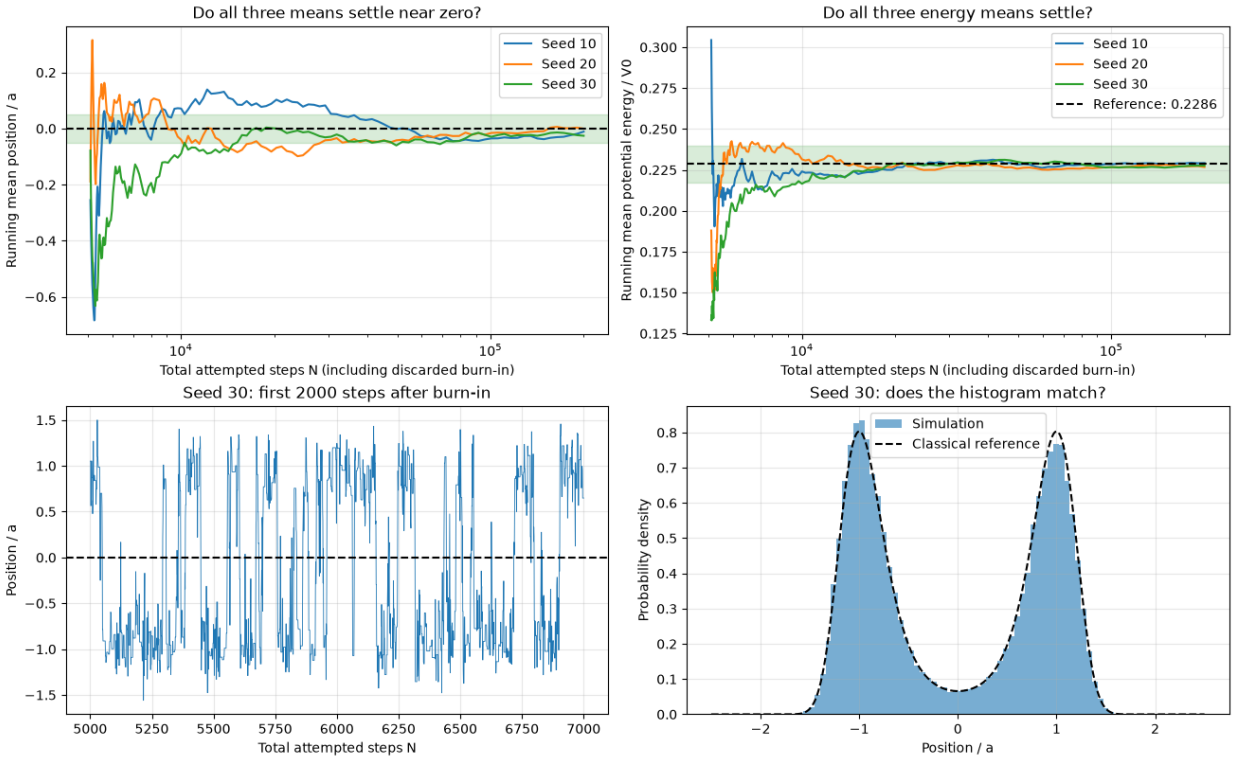


Figure 3.2: Saved running means, a trajectory segment, and a position histogram for the classical validation. The running position mean, rather than the squared centroid position, is monitored.

3.2 Protocols used for the quantum experiments

The production displacement rule is

$$\frac{dx}{a} = 0.4\sqrt{\frac{50}{P}}\,a_{\mathrm{corr}}, \qquad a_{\mathrm{corr}} = 1. \tag{3.2}$$

Thus $dx \simeq 0.730a$ at $P = 15$; the classical calibrated correction factor is not reused unchanged. Final acceptances are approximately 45.6–45.7% and 38.8% for $C = 0.3$ at the two temperatures, and 64.0% and 57.8% for $C = 0.9$. Acceptance alone does not establish good centroid sampling.

Experiment	Discarded sweeps	Production sweeps	Repetitions
P scan, $C = 0.3$, $\theta = 0.4$	5,000	25,000	5 runs
P scan, $C = 0.9$, $\theta = 0.4$	3	29,997	5 runs
Final distributions, $\theta = 0.4$	100,000	400,000	Seeds 10, 20, 30
Final distributions, $\theta = 0.6$	100,000	400,000	Seed 20

Table 3.1: Actual MC schedules. The three-sweep discard in the second P scan is a remaining limitation of that saved experiment.

An earlier 50,000-sweep distribution test at $C = 0.3$, $\theta = 0.4$ discarded 5,000 sweeps. The report uses the subsequent longer experiments. At the lower temperature three centroid densities are averaged before taking their logarithm. The standard deviation of the three individual barrier estimates is reported as run-to-run spread. It is not a standard error or a confidence interval. The higher-temperature cases have no replicated uncertainty estimate. Block averaging and autocorrelation-time analysis have not been performed.

Chapter 4

Deterministic numerical references

4.1 Quantum spectrum and potential energy

In reduced units, the stationary Schrödinger equation reads

$$[-C\partial_y^2 + v(y)]\psi_n(y) = \varepsilon_n \psi_n(y). \tag{4.1}$$

A centered finite difference gives a symmetric tridiagonal matrix with $H_{ii} = 2C/h^2 + v(y_i)$ and $H_{i,i+1} = -C/h^2$. The reference uses 4,001 grid points on $[-4, 4]$, Dirichlet endpoints, and the first 32 eigenstates, calculated with `scipy.linalg.eigh_tridiagonal` [3]. Since numerical eigenvectors obey $\sum_i |u_{in}|^2 = 1$, the density uses $\psi_n(y_i) = u_{in}/\sqrt{h}$.

$$w_n = \frac{e^{-(\varepsilon_n - \varepsilon_0)/\theta}}{\sum_m e^{-(\varepsilon_m - \varepsilon_0)/\theta}}, \quad \rho_q(y) = \sum_n w_n |\psi_n(y)|^2, \quad \langle v \rangle_q = \sum_n w_n \int v(y)|\psi_n(y)|^2 \, dy. \tag{4.2}$$

The MC estimator is compared with $\langle v \rangle_q$, not with the ground-state total energy ε_0. Grid refinement to 8,001 points, enlargement to $[-4.5, 4.5]$ at unchanged spacing, and retention of 48 states change the mean potential energy by less than $5 \times 10^{-7} V_0$ in these four cases. This is a numerical stability check, not an analytical error bound.

C	θ	E_0/V_0	$\langle V \rangle_q / V_0$	$\langle V \rangle_{15} / V_0$	Bias (%)
0.3	0.4	0.738952	0.494056	0.487996	-1.23
0.3	0.6	0.738952	0.497411	0.494576	-0.57
0.9	0.4	1.087697	0.633109	0.611597	-3.40
0.9	0.6	1.087697	0.638393	0.627353	-1.73

Table 4.1: Computed deterministic energies. The final column is the signed finite-$P = 15$ deviation of potential energy from the quantum reference.

The ground-state energies are approximately $0.738952 V_0$ and $1.087697 V_0$, confirming the below-barrier and above-barrier regimes. Figure 4.1 shows how the quantum position density differs from its classical counterpart.

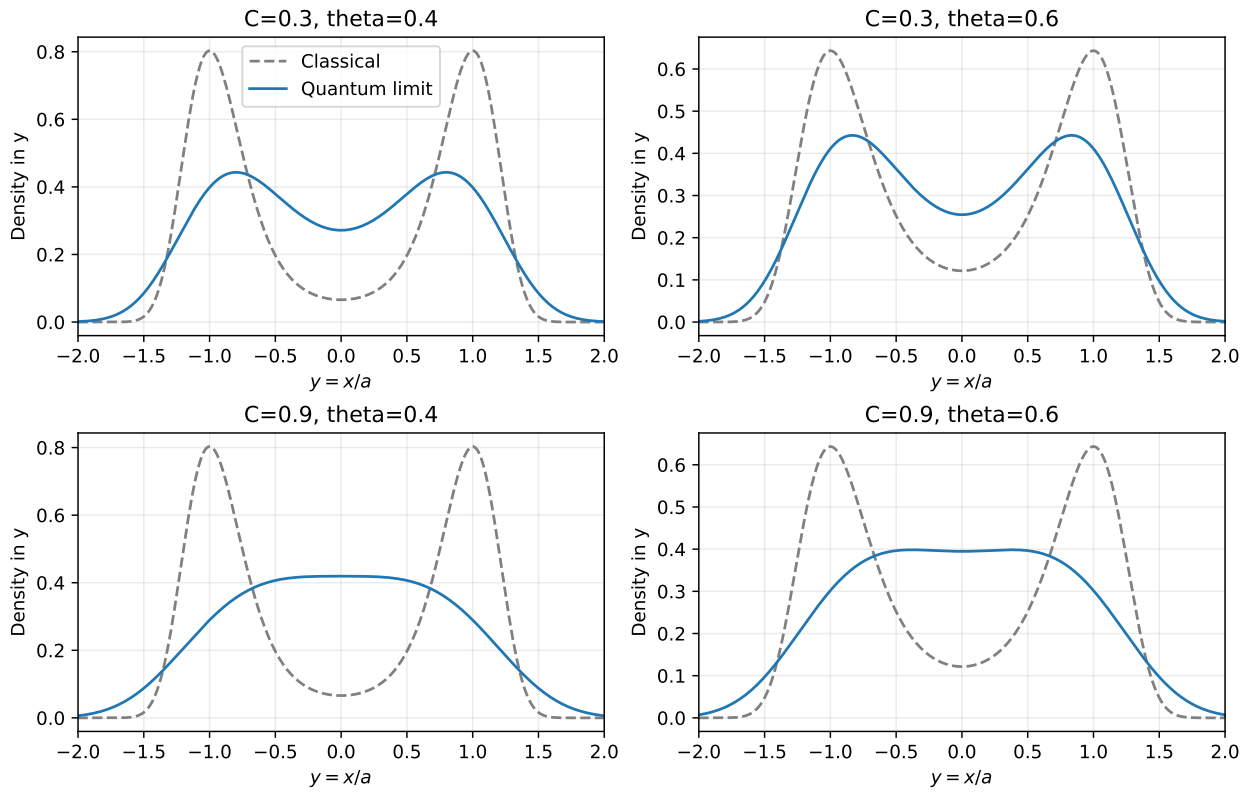


Figure 4.1: Deterministic quantum-limit bead densities and classical reference densities. These curves are independent of the saved MC histograms.

4.2 Transfer matrix for the same finite-P action

The Schrödinger solution benchmarks the quantum limit. To distinguish sampling error from finite-P error, the final notebook section also integrates the action in Eq. (2.4) at the same finite P as the MC. A symmetric quadrature kernel is

$$K_{ij} = h \exp\left[-\frac{P\theta}{4C}(y_i - y_j)^2 - \frac{v(y_i) + v(y_j)}{2P\theta}\right]. \tag{4.3}$$

The omitted free-particle prefactor cancels in normalized expectations. With $Z_P = \mathrm{Tr}(K^P)$, the bead density is $\rho_P(y_i) = (K^P)_{ii}/(hZ_P)$ and $\langle v \rangle_P = \sum_i h\, v(y_i)\rho_P(y_i)$. Matrix rescaling by the largest eigenvalue avoids numerical underflow without changing normalized observables.

For the centroid, define $D(k)_{ii} = e^{iky_i/P}$. Its characteristic function and density are

$$\chi_P(k) = \frac{\mathrm{Tr}[(D(k)K)^P]}{\mathrm{Tr}(K^P)}, \qquad \rho_{c,P}(q) = \frac{1}{2\pi}\int e^{-ikq}\chi_P(k)\, dk. \tag{4.4}$$

The calculation uses 141 quadrature points on $[-3.5, 3.5]$, 80 positive Fourier modes, and period 6.4. Fourier modes are integrated analytically over the same 100 bins as the MC, using the sinc integral of each exponential. References at $P = 15$ and 60 are computed separately.

Doubling the quadrature resolution, increasing the Fourier cutoff to 100 modes, extending the Fourier period to 8, and enlarging the quadrature domain to $[-4, 4]$ change the $P = 15$ binned barrier by less than $2 \times 10^{-5} V_0$. The code also checks normalization and the decay of the characteristic function. Remaining finite-P effects are assessed by comparing $P = 15$ with 60; $P = 60$ is not claimed to be the exact $P \to \infty$ limit.

Chapter 5

Results and comparison

5.1 Convergence with bead count

At $\theta = 0.4$, saved MC scans cover $P = 1$ to 32 for $C = 0.3$ and to 40 for $C = 0.9$. Both show an increasing mean potential energy followed by a noisy plateau. The five individual runs and their mean are plotted, without statistical error bars. The unequal discarded lengths in Table 3.1 and correlated sampling limit the interpretation of that plateau.

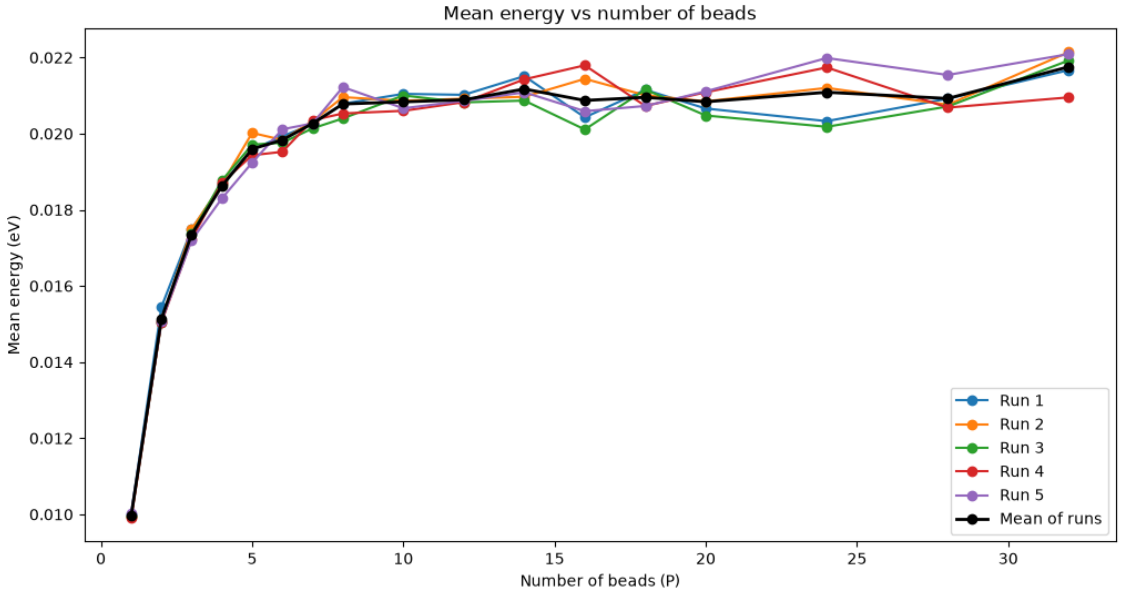


Figure 5.1: Saved MC potential-energy scan at $C = 0.3$, $\theta = 0.4$. The vertical axis is in eV.

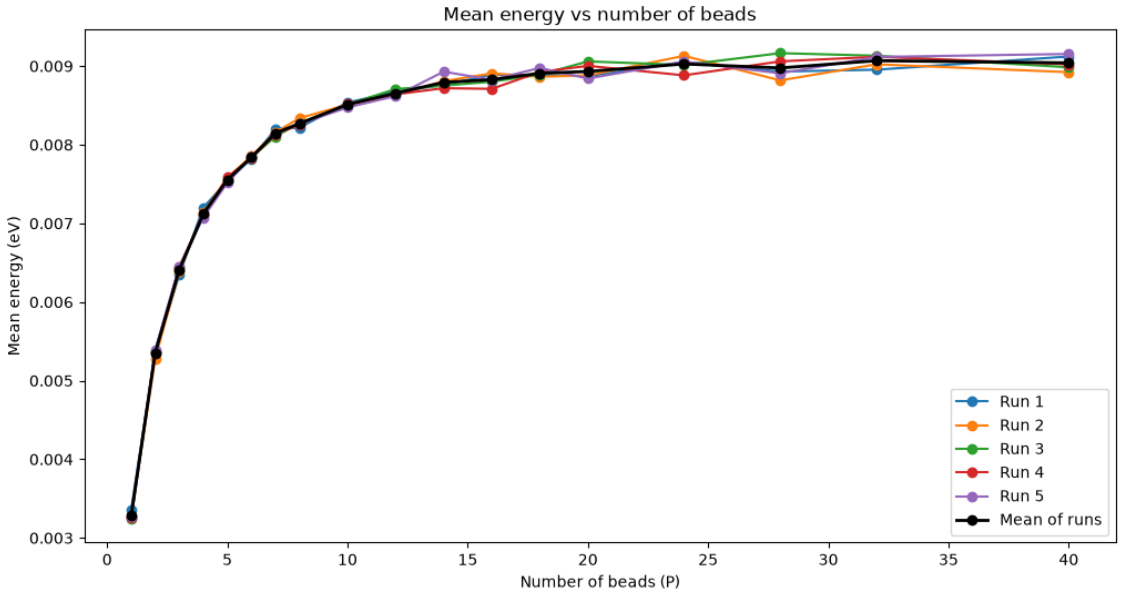


Figure 5.2: Saved MC potential-energy scan at $C = 0.9$, $\theta = 0.4$.

All final MC distributions use $P = 15$ as a practical compromise. It has not been established as an optimal value satisfying a 1% accuracy target. The deterministic finite-P reference gives potential-energy underestimates of about 1.23% for $C = 0.3$ and 3.40% for $C = 0.9$ at $\theta = 0.4$. At $\theta = 0.6$ the corresponding biases are about 0.57% and 1.73%. These discretization biases persist even with perfect MC sampling.

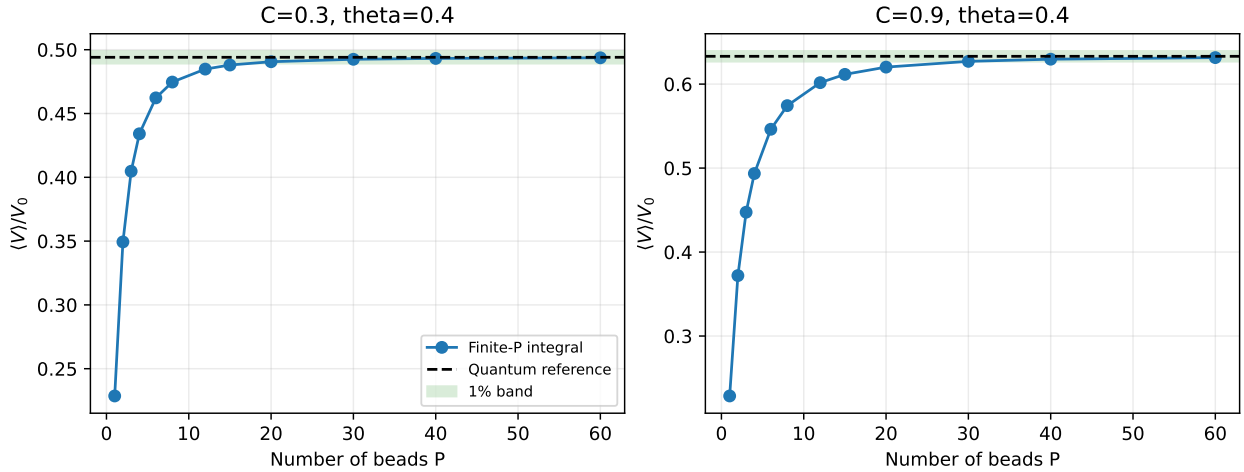


Figure 5.3: Deterministic finite-P potential energies, the Schrödinger reference, and a 1% band. Unlike a noisy MC plateau, this comparison directly isolates the finite-P effect.

The saved MC P scans were not repeated at $\theta = 0.6$, and the measured centroid densities were not compared across P. The deterministic centroid comparison below provides an additional check, but it does not replace those missing MC studies or demonstrate convergence of every observable.

5.2 Bead and centroid distributions

For $C = 0.3$, $\theta = 0.4$, the measured bead density has two broad peaks shifted inward from the classical potential minima and an increased probability near the central barrier. The centroid density

has narrower outer tails and sharper peaks near the wells. These differences reflect fluctuations of the beads about their centroid.

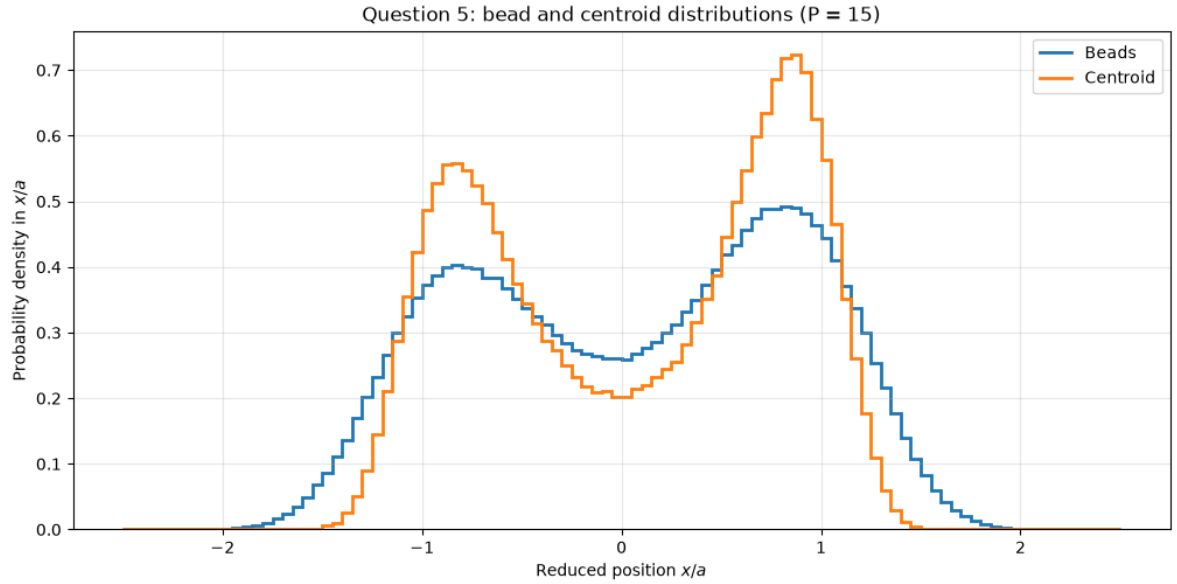


Figure 5.4: Saved $P = 15$ distributions for $C = 0.3$, $\theta = 0.4$, seed 10, after 100,000 discarded and 400,000 production sweeps. This individual run is shown to expose the residual population imbalance.

The unequal peak heights in Figure 5.4 are not a physical breaking of the symmetry: $V(y) = V(-y)$ implies symmetric equilibrium densities. Their variation across seeds is evidence of sampling uncertainty, particularly for the relative populations of the two wells. A normalized histogram area of one does not resolve this problem.

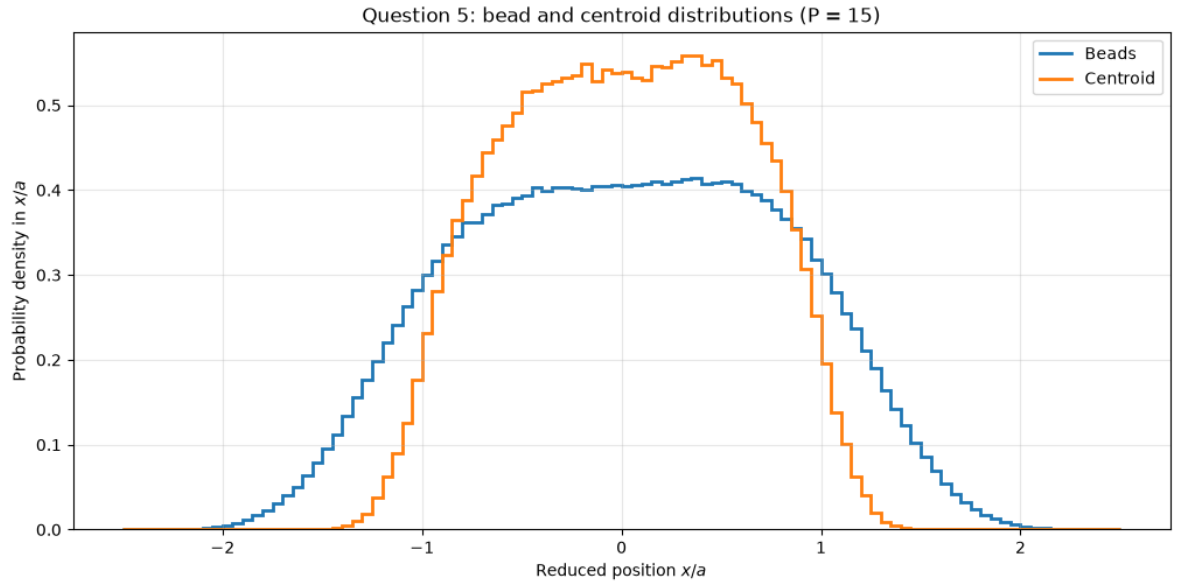


Figure 5.5: Saved $P = 15$ distributions for $C = 0.9$, $\theta = 0.4$, seed 10, with the same sweep schedule.

For $C = 0.9$ the distribution is broader around the origin and the centroid barrier is strongly suppressed. The deterministic quantum-limit bead density has its maximum at the origin at $\theta = 0.4$, while finite-P effects and MC fluctuations can produce a shallow or nearly flat central structure.

At $\theta = 0.6$ the centroid density regains a shallow two-peak shape. The higher-temperature distributions use only one seed per C, so their detailed asymmetry cannot be interpreted as a precise physical effect.

5.3 Effective barriers and their uncertainties

The centroid free-energy profile is defined up to a constant:

$$\frac{F_c(q)}{V_0} = -\theta \ln\left[\frac{\rho_c(q)}{\rho_{c,\max}}\right]. \tag{5.1}$$

For a symmetric bimodal density, the barrier is the difference between the central maximum of F_c and its two equal minima. At $P = 1$, the exact density gives $F_c(y) = V(y)$ and bare barrier V_0.

The implemented finite-bin estimator uses the bin $[0, 0.05]$, centered at 0.025, and the maximum histogram bin density. It approximates the central value, rather than evaluating exactly at zero. With a sampled asymmetric density it also refers to the deeper sampled minimum. Empty histogram bins are masked in the pooled lower-temperature profiles; the existing higher-temperature cells still emit logarithm warnings in empty tails. These tail bins do not enter the reported central estimates.

C	θ	Runs	MC	Run SD	Reference $P = 15$	Reference $P = 60$
0.3	0.4	3	0.458	0.050	0.425299	0.421468
0.3	0.6	1	0.6184	–	0.577325	0.575023
0.9	0.4	3	0.012	0.006	0.000000	0.000000
0.9	0.6	1	0.1395	–	0.098754	0.093603

Table 5.1: Central-bin barriers divided by V_0. Lower-temperature MC values use pooled densities; higher-temperature values use one seed. Run SD is the standard deviation of individual run estimates; a dash means it was not estimated. The deterministic references match the MC bins.

For $C = 0.3$, the pooled lower-temperature barrier is $0.458V_0$, and the three individual estimates are $0.509V_0$, $0.410V_0$, and $0.461V_0$. Their standard deviation is $0.050V_0$, illustrating the remaining statistical variability. For $C = 0.9$, the corresponding values are $0.014V_0$, $0.023V_0$, and $0.012V_0$, with pooled estimate $0.012V_0$ and standard deviation $0.006V_0$. These spreads must not be interpreted as confidence intervals for the pooled estimator.

The matched deterministic $P = 15$ reference is lower than the saved MC barrier in every case. Thus the discrepancy cannot be explained solely by finite-P bias: it is present even when comparing the same discretized path integral and bins. Slow centroid sampling, imbalanced well populations, and selection of the largest noisy density bin are plausible contributors. Quantifying their effects would require additional statistical analysis.

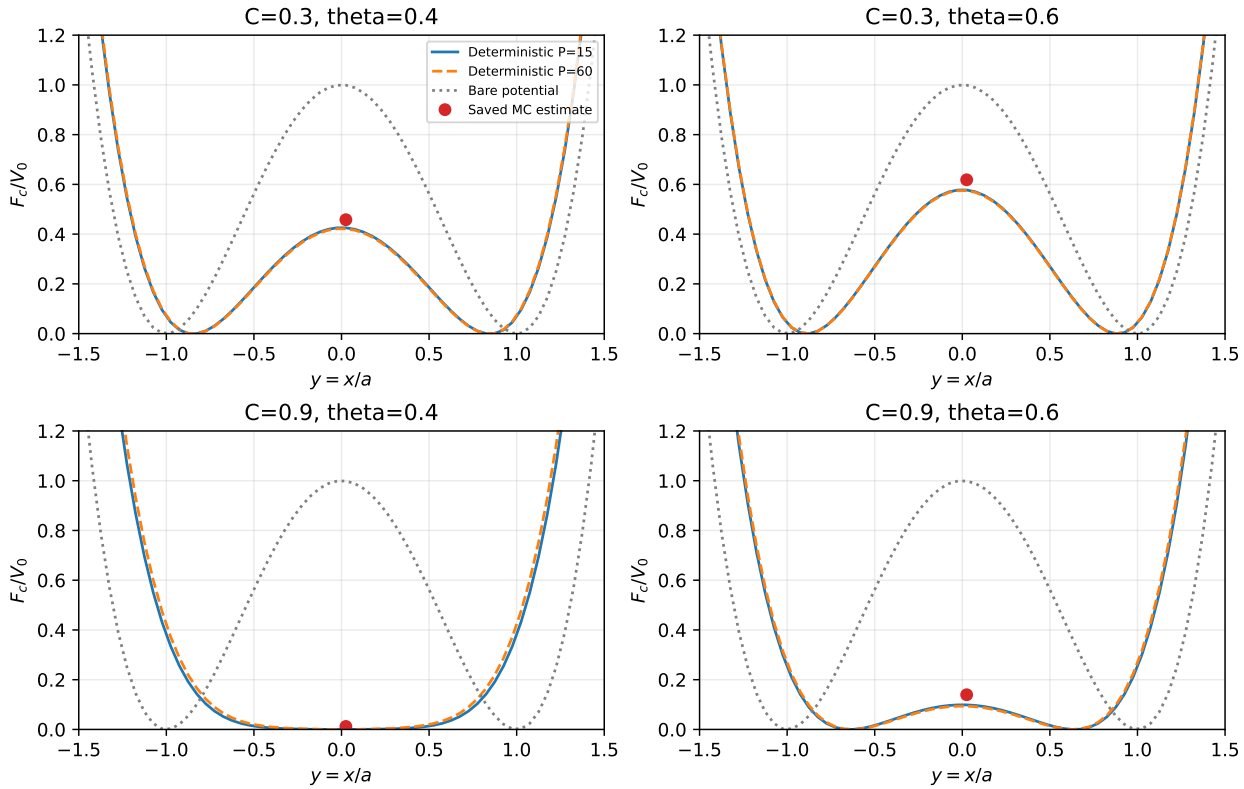


Figure 5.6: Deterministic centroid free energies at $P = 15$ and 60, the bare potential, and the saved MC central-bin estimates. MC markers show estimates without confidence intervals. The references use the same bins as the MC.

The saved barriers increase with temperature for both values of C, while increasing C at fixed reduced temperature reduces them. The deterministic references support these trends. At $C = 0.9$, $\theta = 0.4$, the reference centroid density peaks at the center, so there is no resolved central barrier; the small positive MC value is compatible with a nearly flat profile and sampling noise. At the higher temperature the shallow centroid barrier is restored. These comparisons concern reduced temperatures, whose physical values differ between the two choices of C.

Chapter 6

Discussion and conclusion

The implemented Metropolis updates reproduce the classical distribution and potential-energy reference to the accuracy suggested by three independent runs. The quantum calculations show greater bead delocalization, distinct bead and centroid densities, and an effective barrier substantially below the bare value V_0.

The deterministic section added to the notebook makes the reference values reproducible and separates two errors. The finite-difference Schrödinger solution determines quantum-limit potential energies and bead densities. The transfer matrix integrates the same finite-P path distribution as the MC and supplies a direct benchmark for its centroid estimator. These checks establish that $P = 15$ has measurable discretization bias and that some remaining barrier discrepancies arise within the finite-P sampling problem itself.

The evidence supports a decreasing centroid barrier with increasing C and an increasing barrier with temperature in the cases studied. It does not establish uniform 1% accuracy, an optimal bead count, or confidence intervals for the barriers. Residual asymmetry, a short discarded transient in one bead-count scan, unequal replication across temperatures, and the absence of autocorrelation or block analysis remain limitations. Collective path moves, replicated higher-temperature runs, and observable-specific convergence checks are appropriate next steps. PIMC determines equilibrium quantum statistics; its MC steps and centroid free-energy profiles do not directly give real-time tunneling rates.

Appendix A

Reproducibility and code organization

`main.ipynb` contains the measured experiments and an appended section named "Deterministic references and comparisons." The reference routines are implemented in `src/msn_p8/references.py` using NumPy and SciPy. The final section can be run independently of the long MC cells. It uses summaries retained in the current kernel when available; otherwise it explicitly reads the saved printed MC outputs, at their displayed precision. New MC results should be saved before relying on that fallback.

The deterministic section generates its figures and a machine-readable summary at `output/report/reference_results.json`. The report tables are generated from that summary; original MC figure images are exported from saved notebook outputs. Save newly executed MC cells before refreshing those images. Deterministic energy and barrier values are calculated, not manually entered benchmark constants.

The final reference cells check domain and mesh refinements, thermal-state truncation, characteristic-function decay, and centroid-density normalization. The original MC experiments are retained; adding this reference section does not imply that they were rerun with different parameters or that their sampling errors were eliminated.

Bibliography

[1] H. Dammak, *Modélisation et Simulation Numérique*, Chapter 8, "PIMC applied to a simple 1D system. Double well potential. Tunnel effect (IK)", project brief, pp. 21–22.

[2] D. M. Ceperley, "Path integrals in the theory of condensed helium," *Reviews of Modern Physics* **67**, 279–355 (1995). `https://doi.org/10.1103/RevModPhys.67.279`.

[3] SciPy documentation, `scipy.linalg.eigh_tridiagonal`, `https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.eigh_tridiagonal.html`.